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STUDY PROJECT

Investigating Cyclists' Speed Responses and Steering-Related Simulator Sickness: a CAVE Bicycle Simulator Study

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1 Introduction

Cyclists are among the most vulnerable road users, and the gradual introduction of automated vehicles into urban traffic raises new questions about how these vehicles should communicate with them. Automated shuttle buses stopping at or near cycling infrastructure create frequent close interactions in which a cyclist must decide whether to brake, yield, or pass. External human-machine interfaces (HMIs) — light signals mounted on the vehicle or on the surrounding infrastructure — are one proposed way to support this decision, but their effect on actual riding behaviour is difficult to observe safely in real traffic. Bicycle simulators offer a controlled and repeatable alternative: the same interaction can be presented to many participants under identical conditions while their riding is recorded continuously.

This study project builds on an earlier experiment conducted in a CAVE bicycle simulator, in which cyclists repeatedly encountered an automated shuttle bus at different bus-stop configurations under three HMI conditions (Zafar, 2026). We extended that dataset with an additional round of participant recruitment and analyse the combined material along two complementary lines. The first line examines how cyclists' speed responds to HMI activation, comparing the vehicle-mounted Bus HMI with the Infrastructure HMI (RQ1). The second line examines the relationship between riding behaviour and self-reported simulator sickness: whether handlebar-behaviour measures relate to post-drive sickness ratings, and whether riding behaviour changes over the course of a drive in a way that tracks the rating given afterwards (RQ2a/RQ2b).

The report is structured as follows. Chapter 2 reviews related work and states the research questions and hypotheses. Chapter 3 describes the simulator, the participants, the collected data, and the pre-processing applied to it. Chapter 4 presents the results, followed by discussion, conclusions, and directions for future work.

2 Related work and research questions

2.1 Related work

2.1.1 Effects of road infrastructure on cyclist behaviour

Previous research has shown that road infrastructure can influence cyclists' behavioural responses. Guo et al. (2023) found that protected bicycle lanes received the highest perceived-safety ratings while also being associated with the lowest average cycling speeds. Providing cyclists with dedicated space may also reduce their stress levels. Similarly, Abadi et al. (2019) found that the movement of a truck had the strongest influence on cycling behaviour. A truck pulling out of a loading zone caused cyclists to reduce their speed and increase their lateral distance from the vehicle. Green pavement markings and warning signs also affected cyclists' speed and lateral position.

More recently, Ágoston and Borsos (2026) used a VR-based bicycle simulator to examine cyclists' gaze behaviour across different infrastructure types and levels of automated-vehicle traffic. Fifty participants cycled through 11 scenarios modelled on a 200m urban road segment in Győr. Infrastructure type had the strongest influence on visual behaviour, with separated lanes encouraging broader visual scanning and shared lanes concentrating cyclists' attention on immediate traffic threats. In comparison, pavement colour and the proportion of automated vehicles had relatively limited effects.

2.1.2 Effects of HMI design on cyclist behaviour

The design of HMI signals can also influence how cyclists understand and respond to vehicles. Al Taie et al. (2024) found that cyclists preferred colour-coded signals that could be identified through quick glances. Their outdoor experiment further indicated that large-area displays and animations reinforcing colour changes were easier for cyclists to notice and understand. However, the presence of an HMI does not necessarily result in safer behaviour. Although Kaß et al. (2020) found that an eHMI helped the cyclists to predict the behaviour of the vehicle and make decisions when the vehicle was braking. When the vehicle continued driving, however, the eHMI signal contributed less to the cyclists' behaviour. It didn't lead to obvious cyclists' deceleration or braking actions. These findings show that the behavioural effects of an HMI depend on both its design and its consistency with the vehicle's actual movement.

2.1.3 Trajectory-based analysis of cyclist behaviour

Trajectory data allow cyclists' responses to be examined as a sequence of actions rather than as isolated decisions. Mohammadi et al. (2024) conducted a fixed-base bicycle-simulator experiment. They used cyclists' trajectories to identify when participants stopped pedalling and reached their maximum brake force at an unsignalised intersection in Gothenburg, Sweden.

Mixed-effects models were subsequently used to analyse braking onset and yielding decisions. The results showed that intersection visibility and the difference in time to arrival affected the timing and intensity of cyclists' responses.

Huertas-Leyva et al. (2018) applied an event-based approach to naturalistic cycling data. They extracted 5,092 braking events using bicycle trajectory data to analyse behavioural changes at the individual level when switching cycling tools (from traditional bicycles to electric bicycles). Their study showed that higher cycling speeds were associated with more frequent abrupt braking and greater deceleration, demonstrating how trajectory data can be used to identify safety-relevant behavioural events.

Zafar (2026) investigated cyclists' interactions with an Automated Shuttle Bus (ASB) using the CAVE bicycle simulator at the Technical University of Munich. The experiment included different bus-stop layouts and three HMI conditions: no HMI, a rear-mounted Bus HMI, and an Infrastructure HMI. The study examined cyclists' perceived safety, HMI acceptance and visual attention. Participants perceived the Bus HMI as more readable, while relying more strongly on the familiar Infrastructure HMI. The trajectory examples further indicated that cyclists adjusted their speed and braking behaviour when approaching the bus and the displayed HMI. This study provided the experimental framework and initial dataset for our study, which extends the earlier work through additional participant data and a systematic comparison of cyclists' trajectory-based responses across HMI conditions.

2.1.4 Validity of bicycle simulators

Bicycle simulators provide a controlled and repeatable environment for investigating cyclists' responses to potentially critical traffic interactions. Evidence concerning the validity of bicycle simulators suggests that their performance depends on the behavioural measure being examined. Zheng et al. (2025) examined two configurations of the CAVE-based bicycle simulator through a relative validation study. In a within-subjects experiment, 21 participants followed a reference path containing sudden lateral shifts at fixed speeds, both with and without active steering force feedback. Active steering significantly reduced steering-angle oscillations, indicating smoother handlebar control, whereas no significant difference was found in the resulting trajectory oscillations. This suggests that active steering affects how cyclists operate the handlebar but may be less important when the analysis focuses only on the resulting trajectory. However, the study did not compare simulator behaviour with real-world cycling, and cycling speed was held constant. It therefore does not establish the absolute validity of simulator-derived speed or braking behaviour.

Baldassa et al. (2025) reconstructed a real road environment containing a roundabout and a T-intersection in Padua, Italy, and compared simulator observations with field data. Their results showed good agreement between the simulator and the real-world environment across several kinematic and behavioural indicators, providing evidence of absolute validity for the examined variables. In contrast, O'Hern et al. (2017) used a within-subject design in which

participants completed comparable tasks on a real road and in a bicycle simulator. Although absolute validity was established for several spatial-positioning measures, average cycling speed and speed reduction when approaching intersections demonstrated relative validity. Simulator speed values may therefore not reproduce real-world values exactly. Nevertheless, simulators can still provide a valid basis for examining relative behavioural differences between experimental conditions.

2.2 Research gap and scope of the study

However, the previous trajectory analysis was mainly descriptive and did not systematically quantify cyclists' speed responses around the exact HMI onset or compare these responses between the Bus-HMI and Infrastructure HMI conditions. To address the research gap, the current study conducts an event-based analysis of cyclists' speed profiles around HMI activation. Accordingly, the following research questions are proposed:

RQ1: What differences in cyclists' speed are observed across different HMI types, particularly between Bus-HMI and Infrastructure-HMI?

RQ2a: Do handlebar-behaviour measures relate to post-drive sickness ratings, between riders or within them?

RQ2b: Does riding behaviour change over the course of a drive in a way that tracks the rating given afterwards?

Based on these research questions, the following hypotheses were formulated to guide the analyses.

H1: Cyclists' speeds are affected by different HMI conditions

H2a: Riders' handlebar-steering behaviour relates to their post-drive sickness ratings within riders: deviations from a rider's own typical steering accompany deviations from their own typical rating.

H2b: Steering amplitude increases over the course of a drive, and increases more on drives that receive a worse post-drive sickness rating.

The two research questions are investigated by different authors: RQ1 by Xiaoting Deng and RQ2a/b by Kriti Shrestha. The experimental study, data collection, and descriptive statistics are joint contributions; pre-processing and results subsections specific to one research question are marked (RQ1) or (RQ2) in their headings. A detailed contribution statement is provided at the end of this report.

3 Methodology

The data analysed in this report were initially collected as part of a previous thesis project and were subsequently extended through an additional round of participant recruitment. The newly recruited participants completed the same cycling-simulator experiment, increasing the sample size while maintaining a consistent study design and data-collection procedure. Whereas the previous thesis examined both trajectory and gaze data, the present study focuses on cyclists' trajectory-based behaviour under different HMI conditions, together with participants' self-reported simulator-sickness responses. Video and gaze data were not included because their substantial storage and processing requirements were beyond the scope of the present analysis.

3.1 Simulator and experimental study

3.1.1 CAVE simulator

A Control Automatic Virtual Environment (CAVE) is a virtual-reality installation in which the scene is projected onto large screens surrounding the user, providing visual immersion without a head-mounted display. In the setup used for this study (Zafar, 2026), a stationary physical bicycle stands inside the projection space: the participant's steering and pedalling are transferred to the simulation in real time, so that they control a virtual bicycle through an urban environment containing the automated shuttle bus, other road users, and the HMI installations. The simulator logs the state of the virtual bicycle frame by frame; across the recordings analysed here the effective logging rate was 25–37 Hz (median ≈ 32 Hz), and the bicycle's forward speed was capped by the setup at approximately 7.5 m/s, a limit reached in every recorded drive.



Fig. 3.1 CAVE environment setting

3.1.2 Participants

In the previous thesis work (Zafar, 2026), 41 participants were recruited in a first data-collection wave (February–March 2026); 39 of them completed at least one valid simulator drive, yielding 108 usable drives in our re-audit of the raw material. In the current study project, we enlarged the dataset with a second collection wave (June 2026), adding 9 further participants with 25 usable drives. In total, the analyses in this report draw on 48 participants and 133 drives. 41 of the 48 participants completed all three scheduled drives; the remainder stopped early or lost individual drives to recording failures, including two participants who ended the study after their first drive at the maximum sickness rating of 10 (“vomiting”).

3.1.3 Study design

This study employed a within-subjects design, in which each participant completed three experimental drives corresponding. In each drive, all three HMI conditions will be displayed: no HMI, a vehicle-based Bus HMI, and an Infrastructure HMI. Each drive included the same six bus-stop scenarios, resulting in 18 scenario exposures per participant. The Bus HMI displayed a “STOP” warning on the rear of the automated shuttle when it entered the stopping area, whereas the Infrastructure HMI used yellow and red signals to warn approaching cyclists. The order of the HMI conditions and routes was randomized to reduce potential order and learning effects.



Fig. 3.2 Layout and virtual design of bus stop scenarios

The six scenarios represented different bus-stop locations and infrastructure configurations, including dynamically selected stops with favorable or unfavorable geodata ratings, an existing bus stop, and a novel infrastructure design. The interaction timing was also kept consistent:

the cyclist approached the bus-stop area, the automated shuttle entered the stopping area, and the corresponding HMI changed to its warning state. Tab. 3.1 shows the independent variables.

Variable	Level
Bus stop	Roadside
	Floating bus stop
	Floating bus stop (shared boarding)
Bike infrastructure	None
	Dedicated lane
	Protected lane
eHMI	No HMI
	Bus rear display
	Infrastructure traffic light
Bus type	Automated Shuttle Bus
Traffic	No traffic, only vehicles on the roadside

Tab. 3.1 Independent variables of experimental study

3.2 Data collection

Data used in this study were collected from three sources. The simulator generated time-series trajectory data, including timestamps, positions, and speeds of the cyclist and other relevant road users. Questionnaire data covered participants' demographic characteristics, cycling experience, and simulator-sickness responses. In addition, each experimental drive was recorded by two on-site cameras to document participant behaviour and support the identification of relevant experimental events.

3.2.1 Questionnaire data

Questionnaire responses were collected with a Google form and exported as a spreadsheet for analysis. The export used in this report contains the simulator-sickness responses: each row is one rating instance, identified by a submission timestamp, the participant number, the instance of asking (before the first drive, or after the first, second, or third drive), and the selected Misery Scale (MISC) category as a text label. The MISC is an 11-point scale ranging from "nothing" (0) through increasing dizziness and nausea grades to "vomiting" (10) (Bos et al., 2005). The fields of the export are summarised in Tab. 3.2.

File name	File type	Parameters
MISC (SSQ) [BikeSim2026]	csv	Submission timestamp; participant ID; instance of asking (before first / after first / after second / after third drive); MISC rating as a text label, mapped to the numeric 0-10 scale in pre-processing (Section 3.3.1)

Tab. 3.2 Questionnaire dataset

3.2.2 Trajectory data

Trajectory data were recorded automatically during each simulator ride using a Python-based logging script. Upon completion of each ride, the script generated three CSV files: DynamicObjects.csv, SimBike.csv and StaticObjects.csv. These files contained the recorded simulator parameters (see Tab. 3.3 Overview of the trajectory data files), which were stored on the local computer.

File name	File type	Parameters
DynamicObjects	csv	Timestamp and permutation ID; object name; position $(PosX, PosY, PosZ)$; rotation $(RotX, RotY, RotZ, Ya)$; and object dimensions $(SizeX, SizeY)$ of moving objects, including buses and pedestrians
SimBike	csv	Timestamp and permutation ID; cyclist position $(PosX, PosY, PosZ)$; three-dimensional velocity $(VelX, VelY, VelZ)$; forward velocity $(ForwardVel)$; bicycle orientation and yaw $(RotX, RotY, RotZ, BikeYaw)$; handlebar rotation and yaw $(HandlesRotX, HandlesRotZ, HandlebarYaw)$; bus-station number; and HMI states $(BusHMIState, Infra_HMI_5, Infra_HMI_6)$
StaticObjects	csv	Timestamp and permutation ID; object name; position $(PosX, PosY, PosZ)$; rotation and yaw $(RotX, RotY, RotZ, Yaw)$; and object dimensions $(SizeX, SizeY)$

Tab. 3.3 Overview of the trajectory data files

3.2.3 Video data

Video data is recorded by an on-site camera. However, in this study, we didn't include video data for analysis due to an extreme run time that might be caused by the large data size.

File name	File type	Contents
Date-ParticipantID-PermutationID	mp4	Video taken directly in front of the participant
Date-ParticipantID-PermutationID	mp4	Video taken directly behind the participant

Tab. 3.4 Overview of the video data files

3.3 Data pre-processing

The collected data were pre-processed to derive the variables required for the subsequent analyses. This process included questionnaire preparation, time synchronisation and feature extraction for the necessary calculation and modelling.

3.3.1 Questionnaires

The simulator-sickness responses were processed in three steps. First, the MISC text labels were mapped to their numeric values: “nothing” = 0, “uneasiness” = 1, the four dizziness/warmth/headache grades = 2–5, the three nausea grades = 6–8, “near retching” = 9, and “vomiting” = 10. Second, each rating was linked to the drive it refers to. For the second collection wave this required an empirical clock correction — the questionnaire tool’s clock ran one hour ahead of the simulator clock — after which each post-drive rating was matched to the drive that immediately preceded it; the resulting assignment was verified independently through the scenario-numbering scheme of the trajectory files. Third, irregular entries were resolved: one participant had a duplicate “after first drive” entry stemming from an aborted practice run (the completed block was kept), and four participants’ responses had been entered retrospectively, so their rating values were retained but their timestamps were excluded from all time-based matching.

From the cleaned ratings, per-participant summary variables were derived: the rating after each drive, the peak and mean rating, and the change from the first to the last drive. Participants with fewer than two post-drive ratings were flagged and excluded from analyses that require a within-participant comparison.

3.3.2 HMI events (RQ1)

HMI events were extracted from the state variables recorded in the trajectory data. Tab. 3.5 shows the selection criteria and process. As

Fig. **3.3** shows, a bus HMI event was identified when BusHMIState changed from an inactive state to 1, whereas an infrastructure HMI event was identified at the first active state in the HMI infrastructure from Stop 5 or Stop 6, which is when the state number changed from -1 to others. The corresponding bus stop was obtained from the Infrastructure HMI channel or from the nearest valid BusStationNumber observation for Bus HMI events. After retaining drives containing both Stops 5 and 6, complete ±3s windows, and valid bus-position data, the final dataset contained 156 events from 32 participants: 60 Bus HMI events and 96 Infrastructure HMI events.

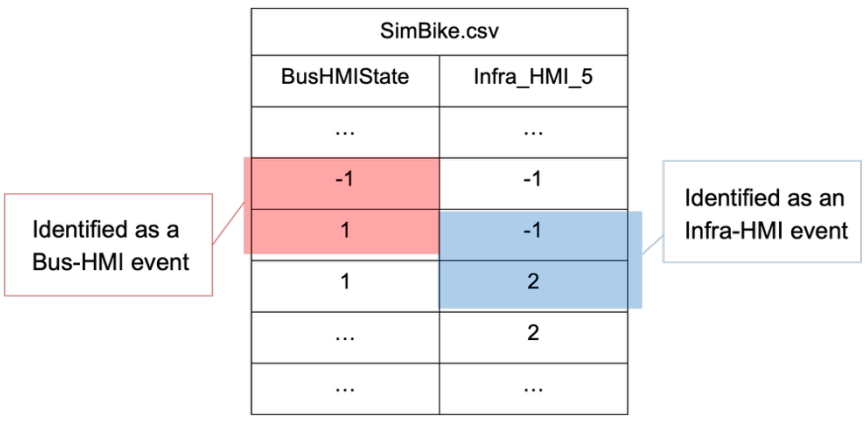


Fig. 3.3 Identify HMI (onset) events from the original dataset

Bus stop No.	HMI condition	Events count	Excluded number of events	Number of events for visualisation and analysis
1	Bus HMI only	31	1[a]	30
2	Bus HMI only	28	/	28
3	Bus HMI only	31	1[a]	30
4	Bus HMI only	28	/	28
5	Bus HMI	32	1[b]	31*
	Infra HMI	87	39[c]	48*
6	Bus HMI	29	/	29*
	Infra HMI	87	39[c]	48*
Total		353	81	272 (For events with * :156)

Tab. 3.5 Selection of HMI events (events marked with an asterisk (*) were included in the Bus vs. Infra HMI effect analysis.)

[a] One Stop-1 event and one Stop 3 event from the same drive were excluded to ensure that the Stop 1-4 visualisations used the same drive set as the final analysis. This drive had no eligible Stop 5/6 events after the bus-position availability check.

[b] One event was excluded because bus-position data were unavailable.

[c] Thirty-nine Infra HMI events were excluded at each stop: one because bus-position data were unavailable and 38 because the corresponding drive did not contain an eligible Bus HMI event.

3.3.3 Cyclists' mean speed (RQ1)

To examine cyclists' short-term speed responses, each HMI event was aligned to its onset time $t = 0$. Raw speed record from 3 seconds before to 3 seconds after HMI onset were extracted and averaged within non-overlapping 0.5s intervals for descriptive visualisation, and then averaged within the pre-3s and post-3s windows for data analysis. No additional smoothing or speed filtering was applied. As some participants contributed multiple events, event-level profiles were first averaged within each participant and subsequently averaged across participants. At Stops 5 and 6, speed profiles were generated for both HMI conditions. From Stops 1 to 4, only speed profiles for Bus-HMI events were available. These profiles were used to visualise and analyse how cyclists' speed responses varied across bus-stop configurations and HMI types.

The 0.5s intervals allow six observations on each period (pre and post) of HMI event, balancing temporal detail and short-term variation. These continuous profiles complement the pre-post comparison by indicating when speed adjustment began and how it developed relative to HMI onset. The use of 3-second window is consistent with Prati et al. (2018), who also evaluated cyclists' mean speeds during the 3s before and after HMI onset.

For the inferential comparison between HMI types, only events at Stops 5 and 6 were included because both Bus-HMI and Infrastructure-HMI observations were available at these locations. Mean bicycle speed was calculated using all raw observations within the complete 3s pre- and post-onset periods, independently of the 0.5s intervals used for visualisation. A linear mixed-effects model (Jashami et al., 2024) was fitted with mean bicycle speed as the dependent variable. Period, HMI type, and bus stop were included as fixed effects, while a participant-level random intercept accounted for repeated observations from the same participant. The interaction between period and HMI type represented the primary test of whether the pre-to-post speed change differed between the two HMI conditions.

3.3.4 Cyclists' peak deceleration location

Peak deceleration was calculated from the cleaned speed trajectory after applying a 0.3s moving-average filter. Acceleration was obtained by differentiating the smoothed speed over time, and the minimum acceleration within the first 3s after HMI onset was defined as the post-onset peak deceleration. The corresponding bicycle-to-bus distance was calculated using the time-aligned bus position. This measure indicates where the cyclist's strongest braking response occurred: a larger distance suggests an earlier response, whereas a smaller distance suggests later braking closer to the bus.

3.3.5 Steering-behaviour metrics (RQ2)

For RQ2, the rider's steering input was isolated from route geometry as the steering deflection: the difference between the handlebar heading and the frame heading, wrapped to ±180°. A deflection of 0° means the handlebars are aligned with the frame, and the measure is independent of where the route points, which makes riders on different scenarios comparable. Recordings were trimmed to sustained riding, only moving frames (speed ≥ 0.5 m/s) were used, and all rates were computed from the recorded timestamps rather than the nominal logging rate. The primary amplitude measure is the interquartile range (IQR) of the steering deflection, which is robust to isolated spikes.

Because drive length differs between the two collection waves (Section 4.1), steering metrics were computed over three 80-second windows at the start, middle, and end of each drive, so that every window contains the same amount of riding. The per-drive quantity carried into the analysis is the within-drive change in steering amplitude: the steering IQR of the last 80

seconds minus that of the first 80 seconds. Drives too short for non-overlapping windows were excluded (131 of the 133 drives qualified).

3.3.6 Within-rider analysis of simulator sickness (RQ2)

The relationship between the within-drive steering change and the post-drive MISC rating was estimated with a linear mixed-effects model over all qualifying drives, with a random intercept per rider. The predictor was person-mean-centred, separating the within-rider component (does a rider rate a drive worse when their steering degraded more than is usual for them?) from the between-rider component (do generally wobblier riders' rate worse overall?). Drive number and collection wave were included as fixed covariates, absorbing practice and accumulation effects and the protocol difference between the waves. The steering-IQR change was declared as the primary endpoint before the model was fitted; the remaining amplitude measures were treated as sensitivity analyses, together with a random-slope variant, a dichotomised-outcome variant, and a first wave-only refit.

4 Results

4.1 Descriptive statistics

The analysed dataset comprises 48 participants and 133 completed drives (Section 3.1.2). Drive length differed between the two collection waves: the median riding time per drive was about 400 s (interquartile range 377-429 s) in the first wave and about 167 s (136-253 s) in the second, which is why analyses that pool the two waves either treat the wave as a covariate or compare time windows of matched length. Mean riding speed was high throughout (on average 7.2 m/s per drive), reflecting the setup's speed cap of approximately 7.5 m/s.

Simulator-sickness ratings covered the full scale but were concentrated at its lower end. Across the 133 drives, the median post-drive MISC rating was 1; 45 drives (34%) were rated 0 ("nothing"), while 15 drives (11%) reached the nausea range (MISC ≥ 6), including three ratings of 10 ("vomiting"). At participant level, 16 of the 48 riders never exceeded MISC 1, 24 peaked in the mild-symptom range (2-5), and 8 reached the nausea range. Ratings were strongly person-dependent: 11 of the 44 riders with at least two rated drives gave the identical rating after every one of their drives, and a further 4 provided only a single rated drive — a compression of within-rider variation that matters for the within-rider analyses of RQ2.

4.2 Cyclists' speed change (RQ1)

summarises the event-level mean speeds during the complete 3s periods before and after HMI onset using all eligible events. Fig. 4.1 presents a sensitivity analysis of these distributions after excluding event-period mean speeds below 2 m/s. While Fig. 4.2 subsequently shows the participant-weighted temporal development of cyclists' speed around HMI onset across all six stops.

Stop	HMI condition	Events	Pre-HMI speed, Mean (SD), m/s	Post-HMI speed, Mean (SD), m/s	Change, Mean (SD), m/s
5	Bus HMI	31	6.906 (1.376)	3.828 (2.524)	−3.078 (2.567)
5	Infra HMI	48	7.373 (0.616)	7.268 (0.973)	−0.105 (0.689)
6	Bus HMI	29	7.497 (0.051)	6.504 (1.659)	−0.993 (1.656)
6	Infra HMI	48	7.375 (0.497)	7.440 (0.343)	+0.065 (0.603)

Tab. 4.1 Descriptive statistics of cyclists' mean speeds during the 3s periods around HMI onset

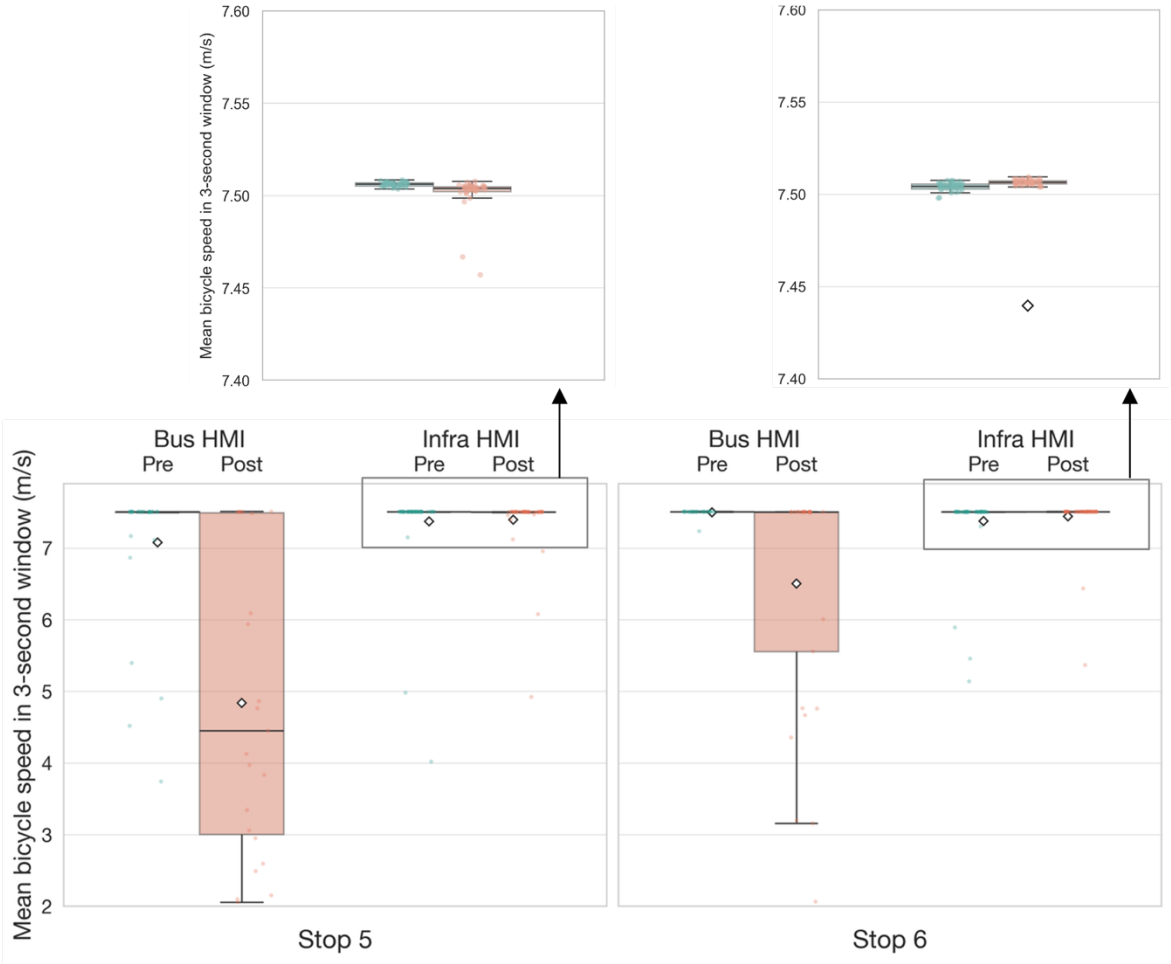


Fig. 4.1 Descriptive statistics on cyclists' mean speed around HMI onset

Fig. 4.1 and Fig. 4.2 present the distinct speed-response patterns across the bus stops and HMI conditions. At Stops 1-3, cyclists' mean speeds were generally between approximately 6.5 and 7.0 m/s before HMI onset. By the end of the observation period, they had decreased to approximately 4.0 m/s at Stop 1, 4.9 m/s at Stop 2, and 5.3 m/s at Stop 3. In contrast, mean speed at Stop 4 remained close to 7.0 m/s, with no clear post-onset reduction.

At Stop 5, mean speed under the Bus HMI decreased from approximately 7.3 m/s at the beginning of the observation window to around 5.8 m/s near HMI onset and 2.6 m/s by the end of the post-onset period. Under the Infrastructure HMI, speed remained around 7.3-7.4 m/s for most of the period and decreased only slightly near the end. At Stop 6, mean speed under the Bus HMI remained close to 7.5 m/s until shortly after onset and subsequently decreased to approximately 5.5 m/s. In contrast, the Infra-HMI profile remained stable at around 7.4 m/s. Figure 4.1 additionally indicates greater variation in post-onset speeds under the Bus HMI, particularly at Stop 5.

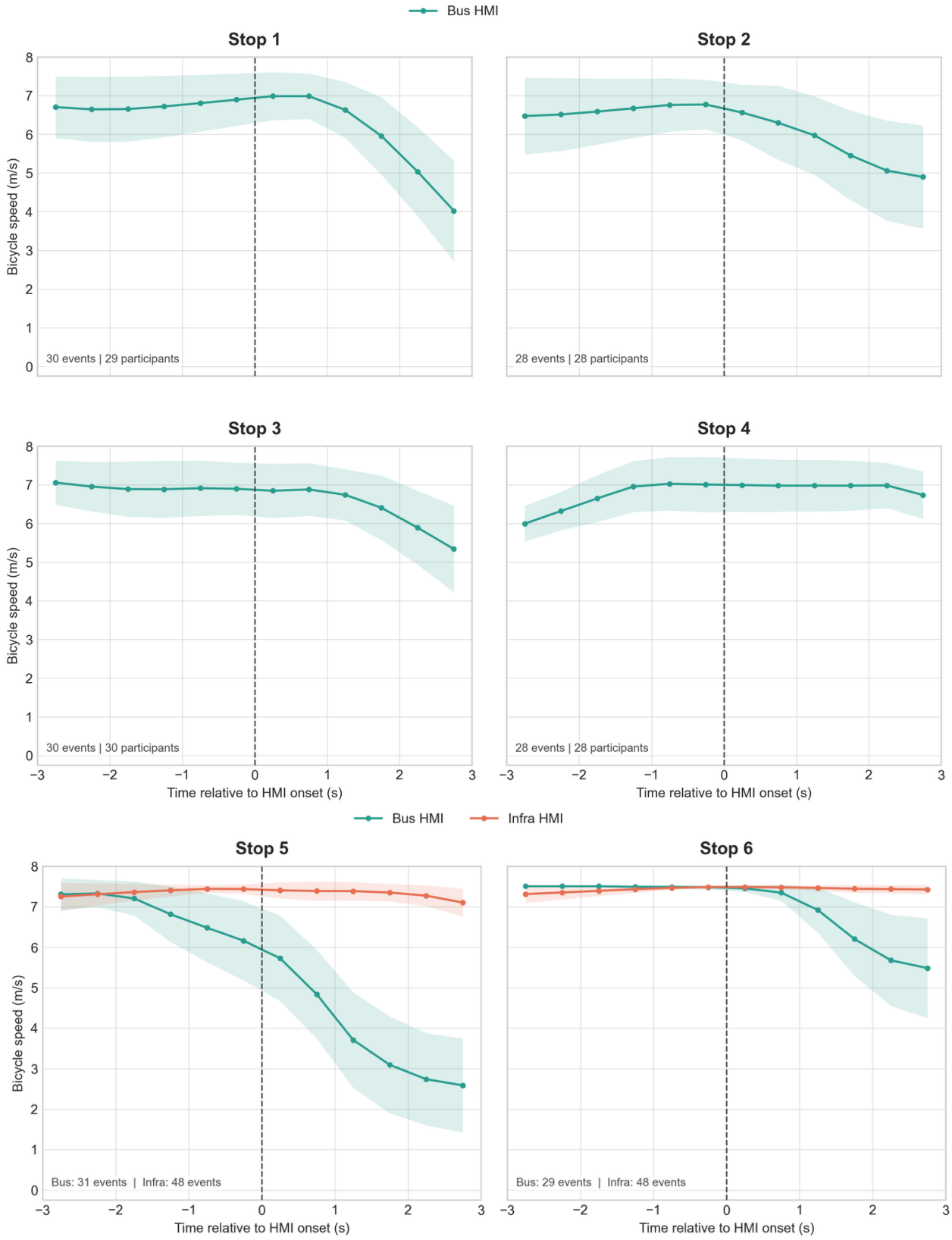


Fig. 4.2 Visualisation of cyclists' mean speed using 0.5s interval during HMI onset

To determine whether the observed pre-to-post speed changes differed between the two HMI conditions, a linear mixed-effects model was fitted. The fixed-effect estimates are presented in Table 4.2.

Fixed effect	Coef.	Std.Err.	z	P	0.025	0.975
Intercept	7.047	0.153	46.155	<.001	6.747	7.346
Period	-0.020	0.176	-0.114	0.909	-0.366	0.326
HMI-type	-0.183	0.202	-0.907	0.364	-0.578	0.212
Station	0.682	0.139	4.920	<.001	0.410	0.953
Period × HMI-type	-2.058	0.284	-7.239	<.001	-2.615	-1.501

Tab. 4.2 Mixed-effects estimates for cyclists' mean speed (RQ1)

The period-by-HMI-type interaction was statistically significant ($\beta = -2.058, p < 0.001$). This indicates that the pre-to-post speed reduction under the Bus HMI was approximately 2.06 m/s greater than under the Infrastructure HMI. Mean speed was also 0.68 m/s higher at Stop 6 than at Stop 5 after accounting for period and HMI type. Neither the period main effect nor the HMI-type main effect was statistically significant.

4.3 Cyclists' peak deceleration location

Tab. 4.3 summarise the distance between the cyclist and the bus at the point of peak deceleration during the 3-second post-HMI period. Fig. 4.3 shows the distribution separated by groups.

For all HMI-onset events, peak deceleration occurred at a mean distance of approximately 20m from the bus. Under the Bus-HMI condition, the median peak-deceleration distance was approximately 13-14m at both Stops 5 and 6. Under the Infra-HMI condition, peak deceleration occurred farther from the bus, with median distances of approximately 19m at Stop 5 and 28m at Stop 6. The difference between the two HMI conditions was particularly apparent at Stop 6.

Variables	Mean	STD	Min	Max
Distance to bus (m)	19.713	7.502	3.864	36.301

Tab. 4.3 Descriptive statistics of numeric variables

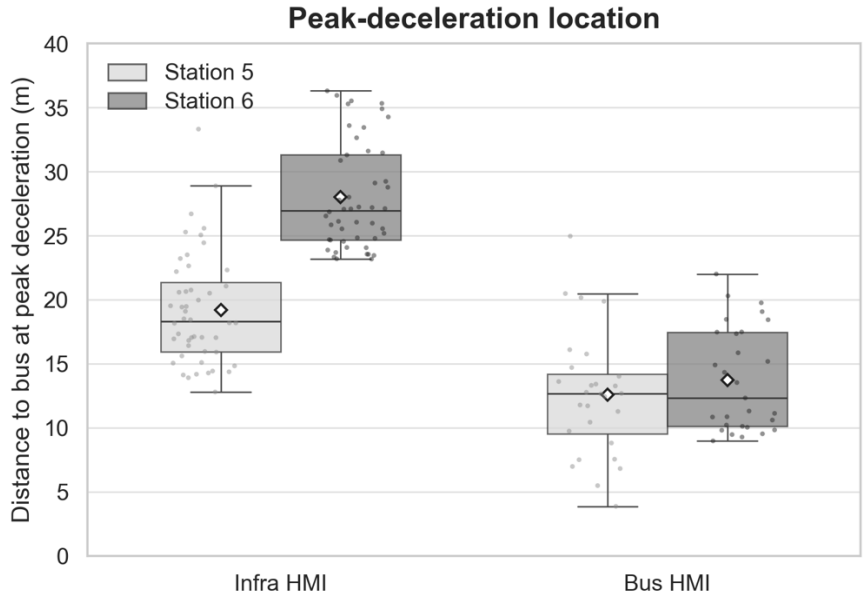


Fig. 4.3 Distribution of cyclists' distance to the bus at post-HMI peak deceleration by HMI condition and bus stop

Overall, the descriptive results suggest that the location of cyclists' strongest speed adjustment varied according to the HMI condition and bus-stop configuration. However, this analysis was exploratory, and no inferential test was conducted. Furthermore, the peak-deceleration location represents the point of greatest observed deceleration within the selected time window rather than a directly observed decision point. The findings should therefore be interpreted as an indication of response patterns rather than evidence of a statistically significant HMI effect.

4.4 Individual differences and the within-drive steering signal (RQ2)

A precondition for RQ2 is knowing how much of the variation in sickness ratings is attributable to the person rather than to the drive. A null mixed-effects model puts the intraclass correlation at 0.69: 69% of all variances in MISC ratings is stable, rider-level susceptibility (consistent with the compression noted in Section 4.1), and only the remaining 31% varies from drive to drive. Between-rider comparisons of steering levels are therefore uninformative about sickness, and all RQ2 analyses compare each rider against their own average.

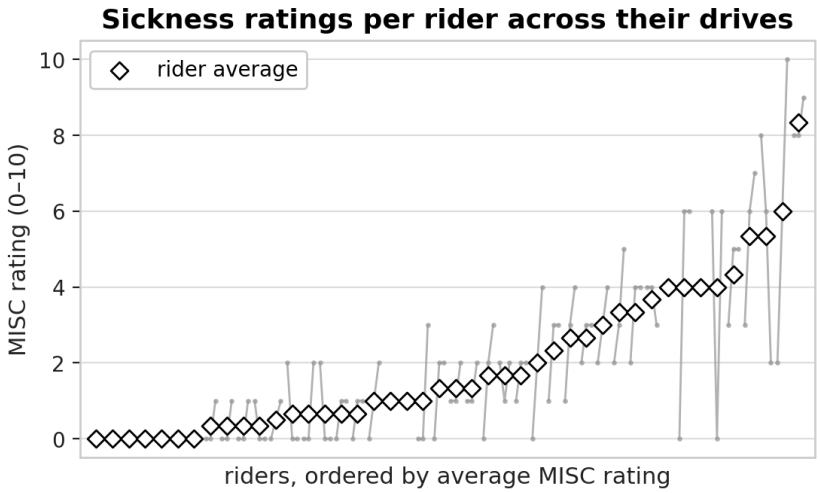


Fig. 4.4 post-drive sickness ratings per rider (grey lines: one rider's drives; diamonds: rider averages)

Within drives, the steering signal emerges only after sickness has had time to build. Across the first collection wave (108 drives, matched drive lengths), steering amplitude in the first 80-second window is uncorrelated with the drive's subsequent rating ($\rho \approx -0.02$, n.s.); in the final window the primary measure correlates at $\rho = +0.24$ ($p < .05$), and related amplitude measures show the same pattern from the middle of the drive onward ($\rho \approx +0.2 \ldots +0.3$). Since all windows are the same length, the contrast reflects position within the drive rather than amount of riding — consistent with sickness accumulating during exposure (H2b).

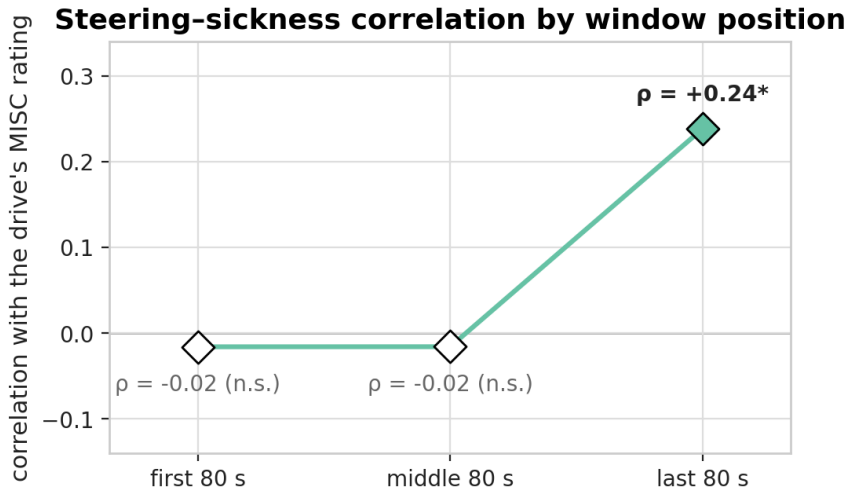


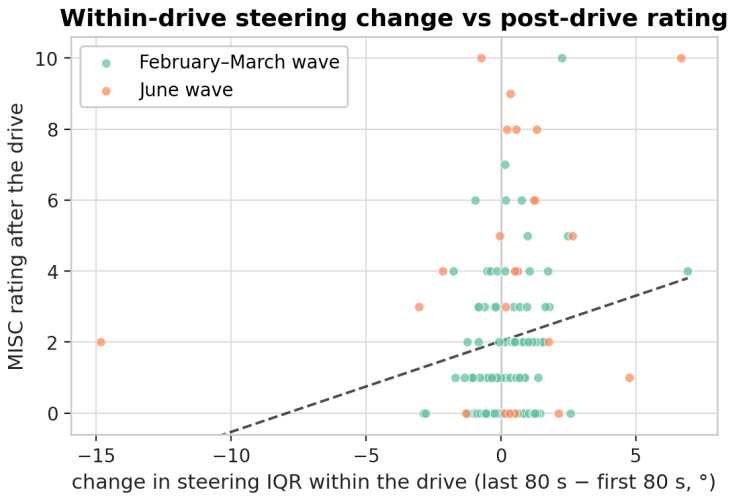
Fig. 4.5 Correlation between steering amplitude and the drive's MISC rating for matched 80-second windows at three positions in the drive

4.5 Mixed-effects model of within-drive steering change (RQ2)

The pre-specified mixed-effects model (131 drives, 48 riders) is summarised in the table below. The within-rider effect is positive and significant: on drives where a rider's steering amplitude grew more than is usual for them, the post-drive rating was higher — by 0.57 MISC points per standard deviation of change — after adjusting for practice, collection wave, and each rider's own susceptibility. This supports H2a and H2b. The model also quantifies the accumulation of sickness across the session (+0.48 MISC per drive) and the difference between the collection waves (+2.89 MISC in the June wave).

Term	Estimate	95% CI	p
Within-rider Δ steering IQR (per SD)	+0.57 MISC	+0.26 … +0.88	< 0.001
Between-rider average Δ steering IQR (per SD)	+0.74 MISC	+0.05 … +1.44	< 0.05
Drive number (accumulation)	+0.48 MISC/drive	+0.21 … +0.75	< 0.001
June collection wave (vs February-March)	+2.89 MISC	+1.36 … +4.42	< 0.001

Tab. 4.4 Mixed-effects model estimates for the within-drive steering change (RQ2)



131 drives, 48 riders; dashed line: linear trend. The mixed-effects model estimates +0.57 MISC per SD of within-rider change (p = .0004).

Fig. 4.6 Within-drive change in steering IQR versus the post-drive MISC rating (131 drives; dashed line: linear trend)

The robustness of this result was probed with four checks, summarised in the table below. Taken together, the effect is reported as supported but not established and should be confirmed on new data (Chapter 7).

Check	Result	Interpretation
Random slope on drive number	Holds (β = +0.41, p < .05)	Not an artefact of assuming one practice trend for all riders
Dichotomised outcome (MISC ≥ 2 / ≥ 3)	Not detectable (OR 1.17 – 1.34, n.s.)	The effect lives in the more severe ratings, not at the mild end
First-wave-only refit	Attenuates (β = +0.23, p > .05)	Partly range restriction: 11 of 44 riders with ≥ 2 rated drives gave the identical rating every time
Other amplitude measures	Same direction, weaker (+0.30 / +0.18 / +0.08)	Only the pre-declared primary measure is significant; disclosed, as the measure was selected from the same data

Tab. 4.5 Sensitivity checks for the within-drive steering effect (RQ2)

5 Discussion

5.1 Cyclists' speed across different HMI types (RQ1)

Regarding RQ1, the results indicate that cyclists' speed responses differed between the two HMI conditions. The significant interaction between period and HMI type showed that the pre-to-post speed reduction under the Bus HMI was approximately 2.0 m/s greater than under the Infrastructure HMI. In contrast, neither the period main effect nor the HMI-type main effect was significant. The findings therefore support H1: cyclists' speeds were not generally lower before HMI onset, but their short-term speed adjustments after HMI onset differed between the conditions.

Although Ahsan (2026) found that some participants were uncertain whether the Bus-HMI signal prohibited overtaking and relied more frequently on the Infra-HMI for decision-making, the present study indicates that cyclists nevertheless responded to this uncertainty by reducing their speed. One possible explanation concerns how cyclists interacted with the two objects. As a dynamic object moving on the road, the ASB may have attracted more attention from cyclists than the fixed infrastructure. Moreover, the rear-mounted STOP sign presented by the Bus HMI may have been interpreted as a direct instruction to slow down or stop, thereby encouraging a more conservative response. Therefore, the present findings reveal a possible distinction between subjective reliance and observable behaviour: an interface perceived as familiar does not necessarily produce the strongest speed response.

At Stop 5, which is located near a road crossing, cyclists began reducing their speed before the Bus-HMI onset. This suggests that not only the presence of the HMI signal but also the station layout may have influenced their behaviour. At Stop 6, where a protected bicycle lane is present, the reduction occurred mainly after onset and was less pronounced. Consequently, the observed difference should be interpreted as the combined effect of HMI presentation, signal timing, and the local traffic context, rather than as an isolated effect of the display location alone.

For the patterns observed in Section 4.3 Cyclists' peak deceleration location, it is possible to be cyclists' habitual reliance on traditional traffic signals. Because the Infra-HMI resembled a familiar traffic light, cyclists may have recognised and responded to it earlier, leading to greater speed adjustments while they were still farther from the bus. In contrast, the Bus-HMI was mounted at a lower height on the rear of the bus and may therefore have become noticeable only at a closer distance. Cyclists may consequently have delayed actions such as slowing down or braking until they were nearer the bus than under the Infra-HMI condition.

5.2 Within-drive steering change and simulator sickness (RQ2)

The results give a consistent answer to RQ2, but a narrower one than the question's original framing. Simulator sickness in this study is first and foremost a property of the person: 69% of all variances in MISC ratings is stable, rider-level susceptibility, and no steering measure distinguishes susceptible from non-susceptible riders when riders are compared with each other. What steering does carry is a trace of sickness as it develops within a person: steering amplitude drifts upward over the course of a drive, and drifts more on the drives that a rider goes on to rate as worse than usual for them (H2a). The timing of the signal supports the same reading — the first 80 seconds of a drive contain no information about the eventual rating, while the later windows do (H2b) — which is consistent with sickness accumulating during exposure rather than being present from the start.

A plausible mechanism is that accumulating sensory conflict between the visual scene and the vestibular system progressively degrades fine motor control, which surfaces as growing, unintentional corrections at the handlebars. This also explains why between-rider comparisons fail: the size of a rider's steering corrections is dominated by individual riding style, so only each rider's deviation from their own baseline is informative. In that respect the RQ2 signal differs in kind from the behaviour studied under RQ1 — the speed adaptations around HMI onset are deliberate responses to a stimulus, whereas the steering drift examined here is an involuntary by-product of the rider's state.

The sensitivity checks (Tab. 4.5) bound the claim. The effect is carried mainly by the more severe ratings and by riders whose sickness actually varied between drives; in the first collection wave, where most riders' ratings barely moved, it attenuates accordingly. The estimate itself is nevertheless robust to the model's main structural assumption (a random slope for practice), and its direction is consistent across all amplitude measures.

6 Conclusion & limitation

With respect to RQ1, we can draw a conclusion that cyclists generally maintained their speed when warning information was provided through the Infrastructure HMI, whereas the rear-mounted Bus HMI prompted a more cautious riding response. Cyclists' reactions can also be shaped by other elements, such as the bus-stop configuration and the surrounding environment. Overall, the findings provide qualified support for H1 and suggest that both the form of external communication and the context in which it is presented are relevant to cyclists' speed adaptation.

This study of RQ1 has several limitations. A considerable number of participants recruited during the second data-collection phase either did not complete the full ride or did not follow the intended route, making their trajectory data unsuitable for the present analysis. This reduced the effective sample for RQ1 to 32 participants and indicates that the experimental procedure did not sufficiently ensure adherence to the predefined route. The exclusion of incomplete drives may also have introduced selection bias, as participants who completed the required scenarios may have been more familiar with cycling simulators or better able to navigate the virtual environment. Furthermore, the analysis was primarily based on trajectory data. Although video and gaze data were collected, they could not be included because of the resources required to process the large files. The observed speed changes could therefore not be directly linked to participants' visual attention, interpretation of the HMI, overtaking decisions or lane choices. Finally, HMI type, activation timing and bus-stop configuration could not be fully separated, meaning that the findings should be interpreted as context-dependent behavioural patterns rather than isolated causal effects of the HMI designs.

Within riders, drives on which steering amplitude degraded more than the rider's own average ended with worse sickness ratings (+0.57 MISC points per standard deviation of change, $p = .0004$), and the steering signal appeared only in the later part of a drive. Both hypotheses are therefore supported in the direction predicted, with the overall verdict supported, not established: the effect should be confirmed on new data before being relied upon. Between riders, steering level carries no information about sickness — the practical implication being that any behavioural monitoring of simulator sickness must be calibrated per rider, not against a population norm.

The main limitations are the following:

– MISC is an ordinal, zero-heavy scale; the linear model treats it as continuous, and the effect is not detectable when the outcome is reduced to a mild/not-mild split.

– The primary outcome measure was selected using the same dataset the model was then fitted to; the model is a sharper test of one hypothesis, not an independent replication.

– The two collection waves differ in drive length and overall sickness severity; the wave covariate absorbs this statistically but cannot fully separate protocol effects from genuine differences.

– A quarter of the riders never changed their rating across their drives, limiting the within-rider variation available to the analysis.

– With at most three drives per rider, the result establishes an association, not a causal mechanism.

7 Future work

Future work should first recruit a larger participant sample and incorporate clearer wayfinding guidance into the virtual environment. Additional signs, road markings, or direct text prompts could provide participants with clearer information about the intended route and increase the chance that they complete the full drive without getting lost. This would reduce the number of unusable trials caused by route deviations and improve the overall efficiency of data collection.

Another possible future work could be comparing laboratory observations with data collected under comparable real-road conditions to assess the behavioural validity of the CAVE bicycle simulator. Following the within-subjects approach applied by O'Hern et al. (2017), participants could complete comparable routes and riding tasks in real-world environments, allowing speed profiles, braking onset, peak deceleration, and trajectory patterns to be compared with the lab data. This would help determine whether the simulator reproduces absolute real-world values or primarily preserves relative differences between experimental conditions. This distinction is important because O'Hern et al. (2017) established relative validity for cyclists' speed measures, while Baldassa et al. (2025) demonstrated absolute validity for selected kinematic and behavioural indicators. Such validation would therefore clarify how confidently the observed HMI-related responses can be interpreted beyond the laboratory and identify outcome measures that may require further simulator calibration.

Several directions follow from the limitations encountered in this project. For the sickness strand, the most important is to collect data in which sickness actually varies within a rider: longer or more provocative drives, or targeted recruitment of more susceptible participants, would give within-rider analyses more to detect — in the present data, a quarter of the riders never changed their rating at all. Asking for a brief MISC rating during each drive, rather than only afterwards, would allow the build-up of sickness to be tested directly instead of being inferred from a comparison of the early and late parts of each recording. Any further data collection should also keep a single, consistent drive protocol: the two existing waves differ in drive length, which had to be controlled for statistically rather than avoided by design. Finally, the within-rider steering result reported here should be confirmed on new data before being treated as established.

For the HMI strand, the event-based speed analysis could be extended to additional bus-stop configurations and to the deceleration profiles of individual approaches, and the recorded video material — excluded here for reasons of processing effort — would allow gaze-based measures to complement the trajectory-based ones.

Author contributions

The contributions of the two authors are listed below, following the CRediT taxonomy extended with categories specific to this project. Sections not attributed to a single research question (experimental study, data collection, descriptive statistics) are joint contributions.

Contribution	Contributor(s)
Conceptualization	X. Deng (RQ1); K. Shrestha (RQ2)
Methodology	X. Deng (RQ1); K. Shrestha (RQ2)
Data collection — second wave (June 2026)	X. Deng and K. Shrestha, jointly
Questionnaire (MISC) pre-processing, participant–drive linkage, and data-quality auditing	K. Shrestha
HMI event extraction and speed/deceleration pipeline	X. Deng
Formal analysis and statistical modelling	X. Deng (RQ1); K. Shrestha (RQ2)
Visualization	X. Deng (RQ1); K. Shrestha (RQ2)
Writing — original draft	Joint sections as described in Section 2.2; each author drafted the sections of their own research question
Writing — review & editing	Both authors

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List of Abbreviations

ASB	Automated Shuttle Buses
CAVE	Control Automatic Virtual Environment
Infra	Infrastructure
HMI	Human-machine Interface

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Appendix A: Questionnaire (Misery Scale / MISC)

The self-reported simulator-sickness responses analysed in Section 3.2.1 and Section 3.3.1 were collected with the following single-item questionnaire, administered via Google Forms before the first drive and again immediately after each drive:

“Please tick the appropriate box according to your condition before and during the simulator study (Misery Scale (MISC)). Note the increasing scale.”

Response options (mapped to the numeric 0–10 MISC scale in Section 3.3.1):

0 – Nothing

1 – Uneasiness

2 – Dizziness, warmth, headache, stomach awareness, sweating – vague

3 – Dizziness, warmth, headache, stomach awareness, sweating – slight

4 – Dizziness, warmth, headache, stomach awareness, sweating – fairly

5 – Dizziness, warmth, headache, stomach awareness, sweating – severe

6 – Nausea – slight

7 – Nausea – fairly

8 – Nausea – severe

9 – Nausea – (near) retching

10 – Vomiting